Practice workbook · Parallel and perpendicular lines: proving it or finding an unknown

Using the rule that parallel lines have equal gradients and perpendicular gradients multiply to -1, either to prove a relationship or to find an unknown coordinate or gradient.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-05

1. Question 5 · part a · 3 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part a only.

5 The points P, Q, R and S have coordinates (4, 7), (3, 0), (10, 1) and (11, 8) respectively.

(a) Show, by calculation, that the lines PR and QS are perpendicular.

(3)

(b) Find the exact lengths of (i) PR (ii) QS

(2)

(c) Find the area of the quadrilateral $PQRS$

(2)

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1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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2. Question 10 · part c.i · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part c.i only.

10 The equation of the line L_1 is $y - 3x + a = 0$

The point A with coordinates $(a, 10)$ lies on L_1

(a) Show that $a = 5$

(2)

Line L_1 crosses the y-axis at the point B

(b) Write down the coordinates of B

(1)

The point C with coordinates $(50, -5)$ lies on L_2

Given that L_2 passes through A

(c) (i) show that L_1 and L_2 are perpendicular

(3)

(ii) hence find an equation for L_2 giving your answer in the form $y = px + q$

(2)

The point D has coordinates (m, n)

The length of CD is $15\sqrt{10}$ and the gradient of BD is -3

(d) Find the value of m and the value of n

(6)

(e) Find the area of quadrilateral $ABCD$

(3)

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2. Question 10 · part c.i · 3 marks · Working and answer

Working and answer

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3. Question 4 · part d.ii · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

4 The point A has coordinates $(-4, -10)$ and the point B has coordinates $(3, 11)$
The line l passes through A and B.

(a) Find an equation of l.

(2)

The point P lies on l such that $AP : PB = 3 : 4$

(b) Find the coordinates of P.

(2)

The point Q with coordinates (m, n), where $m < 0$, lies on the line through P that is perpendicular to l.

Given that the length of PQ is $\sqrt{10}$

(c) find the coordinates of Q.

(6)

The point R has coordinates $(-11, -21)$

(d) Show that

(i) AB and RQ are equal in length,

(ii) AB and RQ are parallel.

(4)

(e) Find the area of the quadrilateral $ABQR$.

(2)

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3. Question 4 · part d.ii · 4 marks · Working and answer

Working and answer

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4. Question 2 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

2 The point A has coordinates $(-7, -1)$ and the point B has coordinates $(3, 4)$

(a) Find an equation of the line that passes through A and B
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(3)

The point C has coordinates $(-3, 7)$

Given that k is a constant such that $AB = kAC$

(b) find the value of k

(2)

The point D has coordinates $(3, p)$ where p is a constant.

Given that CD is perpendicular to AB

(c) find the value of p

(3)

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4. Question 2 · part c · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 8 marks)

4. Question 2 · part c · 3 marks · Working and answer

Working and answer

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5. Question 11 · part b · 7 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

11 The points A and B have coordinates $(-3, -5)$ and $(7, 5)$ respectively.

(a) Find an equation for the line AB

(2)

The point C has coordinates $(p, 1)$ where $p < 0$

Given that AC and BC are perpendicular,

(b) prove that $p = -5$

(7)

The point D, where BCD is a straight line, is such that C divides BD in the ratio $4:3$

(c) Find the coordinates of D

(2)

(d) (i) Find the exact length of AC

(ii) Hence, or otherwise, find the area of triangle ABD

(4)

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5. Question 11 · part b · 7 marks · Working and answer

Working and answer

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6. Question 2 · part c · 8 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

2 The point A has coordinates $(3, 2)$, the point B has coordinates $(8, 3)$ and the point C has coordinates $(4, 7)$

(a) Show that ABC is an isosceles triangle.

(2)

The midpoint of BC is M

(b) Find an equation of the line that passes through A and M

Give your answer in the form $y = mx + c$

(3)

The points A, C and D are collinear such that $AD = kAC \quad (k > 1)$

Given that $\angle ABD = 90^\circ$

(c) find the coordinates of D
Show your working clearly.

(8)

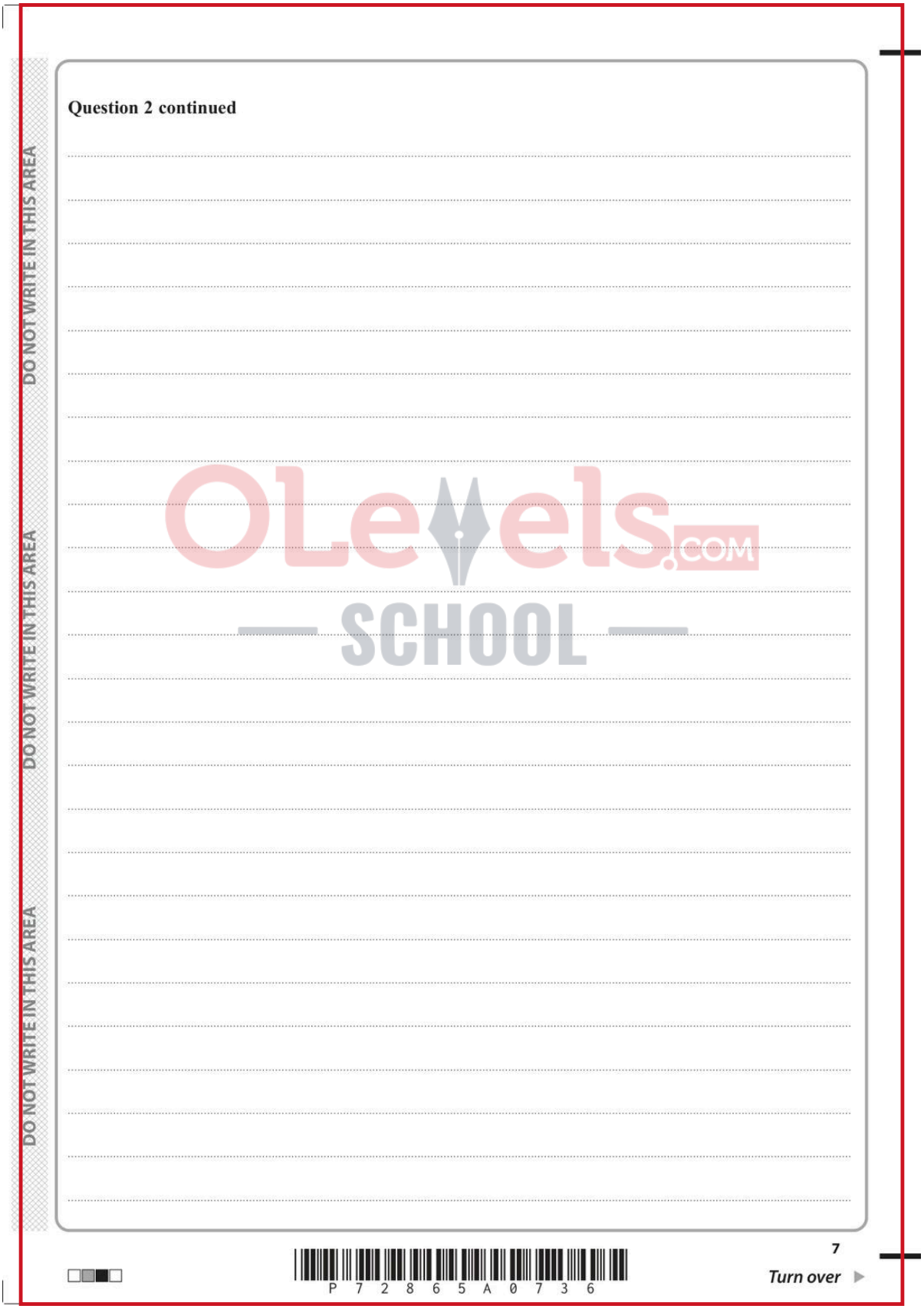
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6. Question 2 · part c · 8 marks

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Further practice · Attempt part c only.



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Question 2 continued

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P 7 2 8 6 5 A 0 7 3 6

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Turn over

6. Question 2 · part c · 8 marks

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Further practice · Attempt part c only.

Question 2 continued

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6. Question 2 · part c · 8 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 13 marks)

6. Question 2 · part c · 8 marks · Working and answer

Working and answer

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7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

9 The straight line L_1 passes through the point A with coordinates (4, 7) and has gradient m, where $m < 0$

Another straight line L_2 is perpendicular to L_1 and passes through the point B with coordinates $(4, k)$ where $k \neq 7$

The lines L_1 and L_2 intersect at the point C.

Given that the y coordinate of C is Y

(a) show that $Y = \dfrac{7 + m^2k}{m^2 + 1}$

(7)

Given that the triangle ABC is isosceles,

(b) find the value of m

(5)

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7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

Question 9 continued

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7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

Question 9 continued

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7. Question 9 · part b · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.



Question 9 continued

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(Total for Question 9 is 12 marks)

7. Question 9 · part b · 5 marks · Working and answer

Working and answer

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